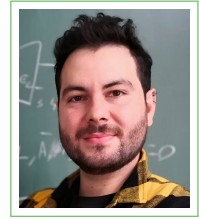


Marco Túlio Quintino

Curriculum Vitae

Associate Professor
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Last update: February 1, 2026

Research Interests

Quantum information and quantum computation, quantum correlations, causality in quantum theory, higher-order quantum operations, Bell nonlocality, EPR steering, entanglement, measurement incompatibility, quantum discrimination tasks, and Semidefinite Programming.

Employment

- Sep 2022 – **Associate Professor (Maître de Conf.)**, Sorbonne University, Paris, France
now Member of the [Quantum Information](#) team at [LIP6](#)
- Mar 2021 – **Postdoctoral researcher**, IQOQI Vienna, Vienna, Austria
- Aug 2022 Postdoctoral Researcher at the group of [Prof. Časlav Brukner](#)
Funding: [ESQ postdoc fellowship](#)
- Jun 2020 – **Postdoctoral researcher**, University of Vienna, Vienna, Austria
- Mar 2021 Postdoctoral Researcher at the group of [Prof. Časlav Brukner](#)
Funding: [Austrian Science Fund \(FWF\)](#) through the [SFB project BeyondC](#)
- Mar 2019 – **Postdoctoral researcher**, The University of Tokyo, Tokyo, Japan
- May 2020 Postdoctoral Researcher at the group of [Prof. Mio Murao](#)
Funding: [Q-LEAP project of the MEXT Japan](#)
- Nov 2016 – **Postdoctoral researcher**, The University of Tokyo, Tokyo, Japan
- Nov 2018 Postdoctoral Researcher at the group of [Prof. Mio Murao](#)
Funding: [JSPS postdoc fellowship](#)

Education

- Jan 2025 **Habilitation in Computer Science [HDR]**, Sorbonne University, Paris, France
Thesis: [Quantum information processing via higher-order operations](#)
Defence date: 17 Jan 2025
Jury: [Pablo Arrighi](#), [Anne Broadbent](#), [Giulio Chiribella](#), [Elham Kashefi](#), [Mário Ziman](#)
- Oct 2012 – **PhD in Physics**, Université de Genève, Geneva, Switzerland
- Sep 2016 Thesis: [Quantum entanglement and measurement incompatibility as resources for nonlocality](#)
Defence date: 09 Sep 2016
Supervisor: [Nicolas Brunner](#)
Funding: [SNF](#), Switzerland

Aug 2010 – **MSc in Physics**, *UFMG*, Belo Horizonte, Brazil
Sep 2012 Thesis: **Black box correlations: locality, noncontextuality, and convex polytopes**
Defence date: 12 Sep 2012
Supervisor: **Marcelo Terra Cunha**
co-supervisor: **Daniel Cavalcanti**
Funding: **CAPES**, Brazil
Feb 2007– **BSc in Physics**, *UFMG*, Belo Horizonte, Brazil
Aug 2010 Monograph: **Não-localidade como recurso para comunicação**
Supervisor: **Marcelo Terra Cunha**
Funding: **CNPq**, Brazil

Grants and awards

- 2025 **Research grant**, **Jeunes Chercheuses et Jeunes Chercheurs (JCJC)**, **ANR**, France
- 2024 **Research grant**, **Programme Polonium 2026**, together with **Michał Studziński**, France and Poland
- 2024 **Research grant**, Tremplins nouveaux entrants & nouvelles entrantes, France
- 2023 **Funding for PhD student**, QuantEdu France (via **PCQT**), France
- 2020 **Postdoc research fellowship**, **ESQ Postdoc Fellowship Program**, Austria/EU
- 2016 **Grant-In-Aid for Scientific Research**, **KAKENHI**, Japan
- 2016 **Postdoc research fellowship**, **JSPS Postdoc Fellowship**, Japan
- 2012 **Master's's research stipend**, **CAPES Master's's Stipend**, Brazil

Languages

Portuguese (native), English (fluent), French (fluent), Spanish (advanced), Japanese (intermediate)
Matlab (fluent), Mathematica (fluent), Julia (elementary), Python (elementary), Bash (elementary)
I use GitHub (<https://github.com/mtcq>) to share relevant computational code I develop.

Publications

Summary: 57 research articles (combining arXiv preprints and peer-reviewed journal publications).
Journal publications include 13 Phys. Rev. Lett., 2 Nat Commun, 1 J. Math. Phys., 1 IEEE Trans. Inf. Theory, and 11 Quantum.

- More than 100 different co-authors.
- **Google Scholar** counts over 2900 citations, h-index of 28, and i10-index of 39 (as of August 2025).
- All my scientific papers are available at **arXiv** and at my **personal website**.
- All publications after I have moved to France may also be found at **HAL**.
- **A list with all my publications can be found at the end of this CV.**

Responsibilities

- Co-responsible (responsible adjoint) of the **Quantum Information master program of Sorbonne University**. This involves participating in the design of the curriculum, participation on the selection process, and participation on the master thesis defence of all students.

Research supervision

Worked in close relation with several students (undergrad, master, and PhD) with different backgrounds at various institutions.

- Concluded the co-supervision of 3 PhD students
- Concluded the co-supervision of 2 master theses
- Concluded the supervision of 4 master internship projects
- Currently supervising 1 PhD student
- Currently supervising 2 master students

PhD supervision

Apr 2025 – **PhD supervision**, *Sorbonne University*, Paris, France

Now Student name: Lucas Porto
Co-supervisor: Mattia Walschaers
Project name: The relationship between quantum entanglement and Bell nonlocality

PhD co-supervision

Oct 2023 – **PhD co-supervision**, *Sorbonne University*, Paris, France

Now Student name: Vanessa Brzić
Main supervisor: Damian Markham
Project name: Higher-Order Quantum Operations: Foundations and Applications

Oct 2023 – **PhD co-supervision**, *Sorbonne University*, Paris, France

Now Student name: Kim Vallée
Main supervisor: Damian Markham
Project name: Bell nonlocality and contextuality

Jun 2017 – **PhD co-supervision**, *UFMG*, Belo Horizonte, Brazil

Jan 2022 Student name: Marcello Nery
Main supervisor: Reinaldo Vienna
Thesis title: Non-classical common-cause and direct-cause
PhD defended on the 21/01/2022

Jan 2018 – **PhD co-supervision**, *The University of Tokyo*, Tokyo, Japan

Dec 2020 Student name: Qingxiuxiong Dong
Main supervisor: Mio Murao
Thesis title: Quantum algorithms for higher-order quantum transformations of universal unitary operations
PhD defended on the 02/2020

Hosting PhD secondments

Mar 2024 – **PhD secondments**, *Sorbonne University*, Paris, France

Avr 2024 Student name: Ties Ohst
Main supervisor: Otfried Gühne
Project title: Quantum memory in causally ordered processes of universal unitary operations

Master's thesis co-supervision

Jul 2018 – **MSc co-supervisor**, *The University of Tokyo*, Tokyo, Japan

Jul 2020 Student name: Wataru Yokojima
Main supervisor: Mio Murao
Thesis title: Consequences of preserving reversibility in quantum superchannels
MsC defended on the 02/2020

Jun 2017 – **MSc co-supervisor**, *The University of Tokyo*, Tokyo, Japan

Jan 2018 Student name: Qingxiuxiong Dong

Main supervisor: Mio Murao

Thesis title: Quantum implementability of maps and supermaps

MSc defended on the 02/2018

Master's internship supervision

Feb 2025 – **Master's internship supervisor**, *Sorbonne University*, Paris, France

Aug 2025 Loic Divry-Kennedy

Project name: Discriminating qubits states and channels from any finite groups

Mar 2025 – **Master's internship supervisor**, *Sorbonne University*, Paris, France

Sep 2025 Charbel Eid

Project name: Optimal discrimination between quantum measurements and quantum instruments

Feb 2024 – **Master's internship supervisor**, *Sorbonne University*, Paris, France

Mar 2024 Yuki Koizumi

Project name: Equivalence between unitary estimation and deterministic port-based teleportation

Jan 2024 – **Master's internship supervisor**, *Sorbonne University*, Paris, France

Apr 2024 Shijun Zhang

Project name: Perfect discrimination of unitary operations when k calls are available

Apr 2023 – **Master's internship supervisor**, *Sorbonne University*, Paris, France

Jun 2023 Shijun Zhang

Project name: Perfect discrimination of unitary operations when k calls are available

Apr 2023 – **Master's internship supervisor**, *Sorbonne University*, Paris, France

Jun 2023 Charbel Eid

Project name: Optimal discrimination between quantum measurements and instruments

Apr 2023 – **Master's internship supervisor**, *Sorbonne University*, Paris, France

Jun 2023 Ethan Obadia

Project name: Quantum channel discrimination with partially entangled states

Participation in examination board

I was a member of the jury (and a reviewer) for 6 PhD defences and 2 MSc defences.

Members of PhD jury

16 May 2025 **Jury of PhD Thesis defence**, *University of Siegen*, Siegen, Germany

Student name: Carlos Belini de Gois

Thesis title: Certifying quantum states and their properties through correlations, communication and tomography

Supervisor: Otfried Gühne

17 Dec 2024 **Jury of PhD Thesis defence**, *Unicamp*, Campinas, Brazil

Student name: Lucas Pollyceno

Thesis title: Multipartite Bell nonlocality and communication

Supervisor: Rafael Rabelo

- 23 Oct 2023 **Jury of PhD Thesis defence**, *Technical University of Denmark Department of Physics*, Copenhagen, Denmark
 Student name: Carles Roch i Carceller
 Thesis title: Quantum state discrimination with applications in contextuality and randomness certification
 Supervisor: Johnatan Bohr Brask
- 12 Jun 2023 **Jury of PhD Thesis defence**, *Université libre de Bruxelles*, Brussels, Belgium
 Student name: Jef Pauwels
 Thesis title: Entanglement and quantum communication between partially characterized devices
 Supervisor: Stefano Pironio
- 10 Mar 2023 **Jury of PhD Thesis defence**, *Unicamp*, Campinas, Brazil
 Student name: Carlos Humberto de Souza Vieira
 Thesis title: Exploring the role of entanglement in sets of behaviors from prepare-and-measure scenarios
 Supervisor: Marcelo Terra Cunha
- 03 Feb 2022 **Jury of PhD Thesis defence**, *Unicamp*, Campinas, Brazil
 Student name: Roberto Dobal Baldijão
 Thesis title: Quantum Darwinism And Contextuality
 Supervisor: Marcelo Terra Cunha
- Members of MSc jury
- 31 Jan 2024 **Jury of MSc Thesis defence**, *UNICAMP*, Campinas, Brazil
 Student name: Arthur Couto Rosa Dutra de Oliveira
 Thesis title: Can KS-Contextuality Hide in a Crowd? Investigating state-independent contextuality in systems with multiple observers.
 Supervisor: Marcelo Terra Cunha
- 09 Apr 2021 **Jury of MSc Thesis defence**, *UNICAMP*, Campinas, Brazil
 Student name: Lucas da S. Pollyceno
 Thesis title: Novos critérios para o princípio da causalidade de informação
 Supervisor: Marcelo Terra Cunha
- Participation on a comité de suivi individuel de thèse (CSI)
- 07 May 2025 **External member of the jury**, *Université Paris-Saclay Alpes*, Palaiseau, France
 Student name: Octave Mestoudjian
 Projet de thèse: Sous-systèmes généralisés en informatique quantique
 Supervisor: Pablo Arrighi
 Co-Supervisor: Augustin Vanrietvelde, Mathew Wilson
- 08 Oct 2024 **External member of the jury**, *Sorbonne University*, Paris, France
 Student name: Youness Ayaita
 Projet de thèse: Le raisonnement ampliatif
 Supervisor: Anouk Barberousse
- 08 Jul 2024 **External member of the jury**, *Université Grenoble Alpes*, Grenoble, France
 Student name: Pierre Pocreau
 Projet de thèse: Implications of causal indefiniteness for quantum communication
 Supervisor: Mehdi Mhalla
 Co-Supervisor: Alastair Abbott

- 01 Jul 2024 **External member of the jury**, *Université Paris-Saclay Alpes*, Palaiseau, France
Student name: Octave Mestoudjian
Projet de thèse: Sous-systèmes généralisés en informatique quantique
Supervisor: Pablo Arrighi
Co-Supervisor: Augustin Vanrietvelde
- 21 Jun 2023 **External member of the jury**, *Université Grenoble Alpes*, Grenoble, France
Student name: Pierre Pocreau
Projet de thèse: Implications of causal indefiniteness for quantum communication
Supervisor: Mehdi Mhalla
Co-Supervisor: Alastair Abbott

Teaching summary

A detailed list of my teaching experience is presented at the end of this CV.

- More than 100h of lectures
- More than 335h of tutorials
- Responsible for creating/reformulating the course “Quantum Circuits and Query Complexity” (previously called “Quantum Circuit and Logic Gates”) for the [Quantum Information master program of Sorbonne University](#).
- Responsible for creating/reformulating the courses “QIIntro” for the [Quantum Information master program of Sorbonne University](#).

Brief description of recent teaching

Undergraduate level in computer science at Sorbonne University:

- Mathématiques Discrètes (TD/TP, 3x)
- Introduction à la calculabilité et à la décidabilité (TD/TP 3x)

Master level in the quantum information master at Sorbonne University:

- Quantum circuits and logic gates (Lecture and tutorials, 2x)
- QIIntro (Lecture and tutorials 2x)
- Quantum kinematics for CS (TD 1x)
- Théorie de Shannon classique et quantique (Cours et TD 2x)

Master level (outside of the quantum domain) at Sorbonne University

- Analyse d’algorithmes et génération aléatoire (TD/TP, 1x)

Outreach and dissemination

During my JSPS postdoc in Japan, I have joined the [JSPS Science Dialogue Program](#), a project where I presented my research in an accessible way to Japanese high-school students. I am also interested in popular science events.

- 21 Sep 2018 [JSPS Science Dialogue](#), Nagano prefecture Suwa-Seiryō High School, Japan
What do you mean by random? Randomness, Quantum Mechanics, and Bell Non-locality
- 05 Dec 2017 [JSPS Science Dialogue](#), Ibaraki prefecture Takezono High School, Japan
What do you mean by random? Randomness, Quantum Mechanics, and Bell Non-locality
- 07 Jan 2013 [Science Slam Berlin](#), SO36, Berlin, Germany
Does God play dice? Genuine randomness in nature

Reviewing activities

Referee for the journals: Quantum, PRL, PRX, PRX Quantum, PRA, PRRResearch, J. Math. Phys, NJP, J. Phys. A, npj Quantum Information, Optica.

I have also refereed submissions for conferences such as QIP, TQC, and AQIS.

Scientific Conferences

- Organised 1 workshop
- Member of the program committee of 4 conferences
- Invited to present a talk in 5 conferences
- Talk accepted in 20 conferences
- Poster accepted in 15 conferences

Organisation

Dec 2023 **Japanese-French Quantum Information 2023 Workshop**, Tokyo, Japan

Program Committee

Oct 2025 **YQIS25**, Barcelona, Spain

Sep 2025 **AQIS25 HK**, The University of Hong Kong (HKU), Hong Kong

Jun 2025 **CEQIP**, Smolenice castle, Slovakia

Sep 2024 **Causalworlds**, Waterloo, Canada

Sep 2024 **AQIS24 Sapporo**, Sapporo, Japan

Sep 2019 **YQIS2019**, Gdansk/Sopot, Poland

Sep 2018 **YQIS2018**, Vienna, Austria

Invited Talk

Sep 2025 **CEQIP**, Smolenice Castle, Slovakia

Talk: Higher-Order Quantum Operations

Jun 2025 **FoQaCiA Workshop 2025**, Braga, Portugal

Talk: Can the quantum switch be deterministically simulated?

Apr 2025 **BQIT:25**, Bristol, UK

Talk: Can the quantum switch be deterministically simulated?

Apr 2022 **Tsirelson Memorial Workshop**, Vienna, Austria

Talk: Measurement incompatibility and Bell nonlocality: from 1985 to 2022

Mar 2020 **The Order of Things (TOOT)**, Obergurgl, Austria

Conference cancelled due to COVID19 pandemic

Jan 2020 **Quantum Information Structure of Spacetime (QISS) 2020**, Hong Kong

Talk: Reversing unknown quantum transformations: A universal quantum circuit for inverting general unitary operations

Nov 2018 **Quantum Maiwar**, Brisbane, Australia

Talk: Semi-device-independent certification of indefinite causal order

Oct 2017 **FQXi Workshop: Quantum Incompatibility**, Laach Lake, Germany

Talk: Genuine n-wise Measurement Incompatibility and Device Independent Certificates of Incompatibility

Accepted Talks

- Jul 2025 **Quantum Physics and Logic 2025 (QPL2025)**, Varna, Bulgaria
Talk: One-to-one correspondence between deterministic port-based teleportation and unitary estimation
- Jul 2025 **Quantum Physics and Logic 2025 (QPL2025)**, Varna, Bulgaria
Talk: All incompatible measurements on qubits lead to multiparticle Bell nonlocality
- Fev 2025 **Quantum Information Processing 2025 (QIP2025)**, Raleigh, United States
Talk: One-to-one Correspondence between Deterministic Port-Based Teleportation and Unitary Estimation
(presented by Satoshi Yoshida)
- Aug 2024 **Asian Quantum Information Science 2024 (AQIS24)**, Sapporo, Japan
Talk: Simulating the quantum switch using causally ordered circuits requires at least an exponential overhead in query complexity
(presented by Satoshi Yoshida)
- Aug 2023 **Asian Quantum Information Science 2023 (AQIS2023)**, Seoul, South Korea
Talk: Simulating qubit correlations with classical communication
- Fev 2023 **Quantum Information Processing 2023 (QIP2023)**, Ghent, Belgium
Talk: Simulating qubit correlations with classical communication
(presented by Martin J. Renner)
- Sep 2021 **SFB BeyondC Autumn Workshop 2021**, Innsbruck, Austria
Talk: Universal protocols for transforming unitary quantum operations
- Jun 2021 **Quantum Physics and Logic 2021 (QPL2021)**, Gdańsk (Online), Poland
Talk: Success-or-draw: A strategy allowing repeat-until-success in quantum computation
- Jan 2020 **Quantum Information Processing 2020 (QIP2020)**, Shenzhen, China
Talk: Adaptive circuits exponentially outperforms parallel ones for universal unitary inversion
- Aug 2019 **Asian Quantum Information Science 2019 (AQIS2019)**, Seoul, South Korea
Talk: Semi-device-independent certification of indefinite causal order
- Sep 2018 **post AQIS18**, Nagoya, Japan
Talk: Reversing unknown quantum transformations
- Sep 2018 **Asian Quantum Information Science 2018 (AQIS2018)**, Nagoya, Japan
Talk: Reversing unknown quantum transformations
- Jul 2018 **Modern Topics in Quantum Information**, Natal, Brazil
Talk: Reversing unknown quantum transformations
- May 2017 **36th Quantum Information Technology Symposium (QIT36)**, Kyoto, Japan
Talk: Super-activation of quantum steering
- Mar 2016 **Quantum Networks (FQXi)**, Barcelona, Spain
Talk: Entangled States With Local Hidden Variable Model For Sequential Measurements
- Dec 2015 **Quantum Correlations, Contextuality and All That... Again**, Natal, Brazil
Talk: Algorithmic construction of local hidden variable models for entangled quantum states
- Aug 2015 **V Quantum Information School and Workshop**, Paraty, Brazil
Talk: Joint measurability, EPR steering, and Bell nonlocality
- Aug 2014 **Asian Quantum Information Science 2014 (AQIS2014)**, Kyoto, Japan
Talk: Joint measurability, EPR steering, and Bell nonlocality

- Dec 2013 **Quantum Correlations, Contextuality and All That**, Natal, Brazil
Talk: Measurement Incompatibility in Quantum Mechanics
- Aug 2013 **IV Quantum Information School and Workshop**, Paraty, Brazil
Talk: Genuine Hidden Quantum Nonlocality
- Jun 2012 **Workshop on Quantum Correlations**, Natal, Brazil
Talk: All noncontextuality inequalities for the n-cycle scenario
- Dez 2011 **III Encontro temático do INCT-IQ**, Natal, Brazil
Talk: From the detection loophole to the transmission loophole
- Aug 2011 **III Quantum Information School and Workshop**, Paraty, Brazil
Talk: Maximal CHSH violations with low efficiency photodetection and homodyne measurements
Poster presentation
- Sep 2021 **Time in quantum theory, ETH workshop**, Zurich (online), Switzerland
Poster: Universal quantum circuits for transforming unitary operations: exponential advantages with causality adaptive strategies and the power of indefinite causality
- Sep 2021 **Vienna Quantum Foundations Conference (VQF-CON 2021)**, Vienna, Austria
Poster: Universal quantum circuits for transforming unitary operations: exponential advantages with causality adaptive strategies and the power of indefinite causality
- Jun 2020 **Theory of Quantum Computation, Communication and Cryptography 2020 (TQC2020)**, Riga (online), Latvia
Poster: Adaptive circuits exponentially outperforms parallel ones for universal unitary inversion
- Dec 2019 **Topical Conference on Quantum Communication and Security 2019 (TCQC2019)**, Kyoto, Japan
Poster: Reversing unknown quantum transformations: A universal quantum circuit for inverting general unitary operations
- Aug 2019 **Asian Quantum Information Science 2019 (AQIS2019)**, Seoul, South Korea
Poster: Distributed sampling, quantum communication witnesses, and measurement incompatibility
- Apr 2018 **International Conference on challenges in Quantum Information Science (CQIS18)**, Tokyo, Japan
Poster: The Cost of Implementing Non-Completely Positive Linear Maps
- Jan 2014 **Quantum Information Processing 2014 (QIP2014)**, Barcelona, Spain
Poster: Genuine Hidden Quantum Nonlocality
- Aug 2013 **IV Quantum Information School and Workshop**, Paraty, Brazil
Poster: Realistic loophole-free Bell test with atom-photon entanglement
- Jan 2013 **Quantum Information Processing 2013 (QIP2013)**, Beijing, China
Poster: Towards a loophole-free Bell test with continuous variables systems
- May 2012 **TQC2012**, Tokyo, Japan
Poster: Perfect homodyne measurements implies CHSH violation with arbitrarily low photodetection efficiency
- Nov 2010 **XIX Semana da Iniciação Científica, UFMG**, Belo Horizonte, Brazil
Poster: Jogo do Quadrado Mágico; Pseudotelepatia Quântica

- Nov 2010 **V Simpósio Nacional / Jornadas de Iniciação Científica**, IMPA, RJ, Brazil
Poster: Não-localidade como recurso para comunicação
- Oct 2010 **WECIQ2010**, Petrópolis, Brazil
Poster: Jogo do Quadrado Mágico; Pseudotelepatia Quântica
- Oct 2009 **XVIII Semana da Iniciação Científica**, UFMG, Belo Horizonte, Brazil
Poster: Algoritmo de Grover – Selected to the top 8%
- Oct 2008 **XVII Semana da Iniciação Científica**, UFMG, Belo Horizonte, Brazil
Poster: Números Inteiros e Criptografia RSA

Participation

- Jun 2025 **Quantum Information (Benasque)**, Benasque, Spain
Workshop without formal talks
- Avr 2024 **QFoundations of Quantum Physics beyond Bell: Celebrating 60 years of Bell's theorem**, Les Diablerets, Switzerland
- Jun 2023 **Quantum Information (Benasque)**, Benasque, Spain
Workshop without formal talks
- Jun 2019 **Quantum Information (Benasque)**, Benasque, Spain
Workshop without formal talks
- Jun 2017 **Quantum Information (Benasque)**, Benasque, Spain
Workshop without formal talks
- Jun 2015 **Quantum Information (Benasque)**, Benasque, Spain
Workshop without formal talks
- Aug 2015 **V Quantum Information School and Workshop**, Paraty, Brazil
- Sep 2015 **Gisin's 60th birthday workshop**, Riederalp, Switzerland
- Jun 2013 **Quantum Information (Benasque)**, Benasque, Spain
Workshop without formal talks
- Aug 2013 **IV Quantum Information School**, Paraty, Brazil
- Jul 2012 **62nd Lindau Nobel Laureate Meeting dedicated to Physics**, Lindau, Germany
- Aug 2011 **III Quantum Information School and Workshop**, Paraty, Brazil
- Jul 2011 **28o Colóquio Brasileiro de Matemática**, IMPA, Rio de Janeiro, Brazil
- Aug 2010 **XIV Escola Brasileira de Probabilidade**, Búzios, Brazil
- Jul 2010 **Clay Mathematics Institute 2010 Summer School**, *Probability and Statistical Physics in Two and more Dimensions*, Búzios, Brazil

Academic visit seminars

- 30 Apr 2025 **University of Siegen**, Siegen, Germany
Talk: Can the quantum switch be deterministically simulated?
Host: Otfried Gühne
- 12 Dec 2024 **Hong Kong University (HKU)**, Hong Kong
Talk: Can the quantum switch be deterministically simulated?
Host: Giulio Chiribella

- 11 Nov 2024 **University of Geneva**, Geneva, Switzerland
Talk: All incompatible measurements on qubits lead to multiparticle Bell nonlocality
Host: Nicolas Brunner
- 08 Jul 2024 **Université Grenoble Alpes**, Grenoble, France
Talk: All incompatible measurements on qubits lead to multiparticle Bell nonlocality
Host: Alastair Abbott
- 25 Jun 2024 **University of Pisa**, Pisa, Italy
Talk: The relationship between measurement incompatibility and Bell nonlocality
Host: Costantino Budroni
- 24 Oct 2023 **Technical University of Denmark, QPIT**, Copenhagen, Denmark
Talk: Transforming and discriminating quantum operations using higher-order methods
Host: Johnatan Bohr Brask
- 16 Mar 2023 **Les Atelier du LKB, Laboratoire Kastler Brossel**, Paris, France
Talk: Parallel, sequential, and non-causal strategies for transforming unitary operations and discriminating quantum channel via a higher-order approach.
Host: Nancy Paul
- 10 Dec 2021 **University of Gdańsk**, Gdańsk, Poland
Talk: Transforming unitary operations via quantum circuits: Universal unitary inversion, transposition, and complex conjugation
Host: Michał Studziński
- 18 Mar 2021 **Quantin research group**, Warsaw (online), Poland
Talk: Universal protocols for transforming unitary quantum operations
Host: Michał Oszmaniec
- 05 Feb 2021 **Technical University of Denmark**, Copenhagen (online), Denmark
Talk: Reversing unknown quantum transformations: A universal protocol for inverting general unitary operations
Host: Jonatan Bohr Brask
- 10 Jul 2019 **Universitat Autònoma de Barcelona**, Barcelona, Spain
Talk: Reversing unknown quantum transformations: A universal protocol for inverting general unitary operations
Host: Andreas Winter
- 14 Feb 2019 **ICFO**, Barcelona, Spain
Talk: Reversing unknown quantum transformations: A universal protocol for inverting general unitary operations
Host: Antonio Ácin
- 12 Feb 2019 **GAP**, Geneva, Switzerland
Talk: Reversing unknown quantum transformations: A universal protocol for inverting general unitary operations
Host: Nicolas Brunner
- 24 Jan 2019 **IQOQI Vienna**, Vienna, Austria
Talk: Reversing unknown quantum transformations: A universal protocol for inverting general unitary operations
Host: Marcus Huber

- 16 Dec 2016 **The University of Tokyo**, Tokyo, Japan
Talk: Super-Activation of Quantum Steering
Host: Mio Murao
- 18 Nov 2016 **UFMG**, Belo Horizonte, Brazil
Talk: Super-Activation of Quantum Steering
Host: Marcelo Terra Cunha
- 18 Nov 2015 **University of Siegen**, Siegen, Germany
Talk: Inequivalence of Entanglement, Steering, and Bell Nonlocality For General Measurements
Host: Otfried Gühne
- 18 Aug 2015 **UFMG**, Belo Horizonte, Brazil
Talk: Inequivalence of Entanglement, Steering, and Bell Nonlocality For General Measurements
Host: Marcelo Terra Cunha
- 31 Jul 2015 **UFMG**, BH, Brazil
Talk: Joint Measurability, EPR Steering, and Bell Nonlocality
Host: Marcelo Terra Cunha
- 15 Jun 2015 **Waseda University**, Tokyo, Japan
Talk: Joint Measurability, EPR Steering, and Bell Nonlocality
Host: Kazuya Yuasa
- 26 May 2015 **PI**, Waterloo, Canada
Talk: Joint Measurability, EPR Steering, and Bell Nonlocality
Host: Matthew Pusey
- 09 Oct 2014 **ICFO**, Barcelona, Spain
Talk: Joint Measurability, EPR Steering, and Bell Nonlocality
Host: Antonio Ácin
- 05 Sep 2014 **KCIK**, Gdansk, Poland
Talk: Joint Measurability, EPR Steering, and Bell Nonlocality
Host: Michał Horodecki
- 25 Aug 2014 **The University of Tokyo**, Tokyo, Japan
Talk: Joint Measurability, EPR Steering, and Bell Nonlocality
Host: Mio Murao
- 03 Dec 2013 **UFMG**, Belo Horizonte, Brazil
Genuine Hidden Quantum Nonlocality
Host: Marcelo Terra Cunha
- 19 Jul 2012 **KCIK**, Gdansk, Poland
Talk: Characterization of the n-Cycle Noncontextual Polytope
Host: Michał Horodecki
- 11 Jul 2012 **ICFO**, Barcelona, Spain
Talk: Characterization of the n-Cycle Noncontextual Polytope
Host: Antonio Ácin

List of publications

Peer-reviewed publications

- [1] S. Egelhaaf, J. Pauwels, M.T. Quintino, and R. Uola. "Certifying measurement incompatibility in prepare-and-measure and Bell scenarios". *J. Phys. A* **58** 095304 (2025).
- [2] M. Plávala, O. Gühne, and M.T. Quintino. "All Incompatible Measurements on Qubits Lead to Multiparticle Bell Nonlocality". *Phys. Rev. Lett.* **134** (2025).
- [3] L. Villegas-Aguilar, E. Polino, F. Ghafari, M. T. Quintino, K. T. Laverick, I. R. Berkman, S. Rogge, L. K. Shalm, N. Tischler, E. G. Cavalcanti, S. Slussarenko, and G. J. Pryde. "Nonlocality activation in a photonic quantum network". *Nature Communications* **15**, 3112 3112 (2024).
- [4] P. Taranto, M. T. Quintino, M. Murao, and S. Milz. "Characterising the Hierarchy of Multi-time Quantum Processes with Classical Memory". *Quantum* **8** 1328 (2024).
- [5] M. Antesberger, M.T. Quintino, P. Walther, and L. A. Rozema. "Higher-Order Process Matrix Tomography of a Passively-Stable Quantum Switch". *PRX Quantum* **5**, 010325 010325 (2024).
- [6] S. Milz and M.T. Quintino. "Characterising transformations between quantum objects, 'completeness' of quantum properties, and transformations without a fixed causal order". *Quantum* **8** 1415 (2024).
- [7] T. Strömberg, P. Schiansky, R.W. Peterson, M.T. Quintino, and P. Walther. "Demonstration of a quantum SWITCH in a Sagnac configuration". *Phys. Rev. Lett.* **131** 060803 (2023).
- [8] T. Strömberg, P. Schiansky, M.T. Quintino, M. Antesberger, L. A. Rozema, I. Agresti, Č. Brukner, and P. Walther. "Experimental superposition of a quantum evolution with its time reverse". *Phys. Rev. Research* **6**, 023071 023071 (2024).
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Teaching details

- 2025/2 **Lecture (20h), Tutorial (6h)**, Sorbonne University, Master's d'Informatique - Information Quantique (IQ), M1, Paris, France
MU4INQ05 - Quantum circuits and logic gates
- 2025/1 **Lecture (20h)**, Sorbonne University, Master's d'Informatique - Information Quantique (IQ), M1, Paris, France
MU4INQ51 - QIIntro
- 2025/1 **Tutorial (36.5h)**, Sorbonne University, Lincese d'Informatique, L3, Paris, France
LU3IN030 - Introduction à la calculabilité et à la décidabilité
- 2024/2 **Lecture (4h), Tutorial (2h)**, Sorbonne University, Master's d'Informatique - Information Quantique (IQ), M2, Paris, France
MU5PYQ03 - QIT: Quantum Information Theory

- 2024/2 **Lecture (10h), Tutorial (8h)**, *Sorbonne University, Master's d'Informatique - Information Quantique (IQ)*, M1, Paris, France
MU4INQ05 - Quantum circuits and logic gates
- 2024/2 **Tutorial (38.5h)**, *Sorbonne University, Lincese d'Informatique, L2*, Paris, France
LU2IN005 - Mathématiques Discrètes
- 2024/1 **Lecture (2h)**, *4EU+ European University Alliance*, University of Copenhagen (online), Denmark
4EU+: Quantum Information and Quantum Many-Body Theory
- 2024/1 **Lecture (10h), Tutorial (6h)**, *Sorbonne University, Master's d'Informatique - Information Quantique (IQ)*, M1, Paris, France
MU4INQ51 - QIIntro
- 2024/1 **Lecture (18h), Tutorial (6h)**, *Sorbonne University, Master's Sciences, Technologies, Santé, M1.*, Paris, France
MU4PY223 - Théorie de Shannon classique et quantique
- 2024/1 **Tutorial (36.5h)**, *Sorbonne University, Lincese d'Informatique, L3*, Paris, France
LU3IN030 - Introduction à la calculabilité et à la décidabilité
- 2023/2 **Lecture (4h), Tutorial (2h)**, *Sorbonne University, Master's d'Informatique - Information Quantique (IQ)*, M2, Paris, France
MU5PYQ03 - QIT: Quantum Information Theory
- 2023/2 **Lecture (10h, Tutorial (6h))**, *Sorbonne University, Master's d'Informatique - Information Quantique (IQ)*, M1, Paris, France
MU4INQ05 - Quantum circuits and logic gates
- 2023/2 **Tutorial (38.5h)**, *Sorbonne University, Lincese d'Informatique, L2*, Paris, France
LU2IN005 - Mathématiques Discrètes
- 2023/1 **Lecture (15h), Tutorial (15h)**, *Sorbonne University, Master's Sciences, Technologies, Santé, M1.*, Paris, France
MU4PY223 - Théorie de Shannon classique et quantique
- 2023/1 **Tutorial (36.5h)**, *Sorbonne University, Lincese d'Informatique, L3*, Paris, France
LU3IN030 - Introduction à la calculabilité et à la décidabilité
- 2022/2 **Tutorial (36.5h)**, *Sorbonne University, Lincese d'Informatique, L2*, Paris, France
LU2IN005 - Mathématiques Discrètes
- 2022/2 **Lecture (2h), Tutorial (16h)**, *Sorbonne University, Master's d'Informatique - Information Quantique (IQ)*, M1., Paris, France
MU4INQ01 - Quantum kinematics for computer scientists
- 2022/2 **Tutorial (14h)**, *Sorbonne University, Master's d'Informatique - Parcours Science et Technologie du Logiciel (STL)*, M2, Paris, France
MU5IN550 - Analyse d'algorithmes et génération aléatoire
- 2022/2 **Lecture (6h), Tutorial (4h)**, *Sorbonne University, Master's d'Informatique - Information Quantique (IQ)*, M2, Paris, France
MU5PYQ03 - QIT: Quantum Information Theory
- 2021 **One lecture**, *Semana da Física 2021 – UNESP*, Caratinguetá (online), Brazil
Não-localidade de Bell: como o indeterminismo quântico permite correlações supra-clássicas

- 2019 **One lecture**, *The University of Tokyo*, Tokyo, Japan
What do you mean by random? Randomness, Quantum Mechanics, and Bell Non-locality – Undergraduate course in quantum technology
- 2016/1 **Teaching Assistant**, *Université de Genève*, Geneva, Switzerland
Quantum Information Theory – Master's Course
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Méthodes mathématiques pour physiciens I – Undergraduate Course
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Quantum Information Theory – Master's Course
- 2011/1 **Teaching Assistant**, *IMPA*, Rio de Janeiro, Brazil
Mecânica quântica para matemáticos em formação – 28o Colóquio Brasileiro de Matemática
- 2011 **One lecture**, *UFMG*, Belo Horizonte, Brazil
Comunicação via qubits – XXII Escola de Inverno
- 2010 **One lecture**, *UFMG*, Belo Horizonte, Brazil
Desigualdades de Bell, uma introdução a não-localidade quântica – Quantum Mechanics Graduate Course
- 2010 **One lecture**, *UFMG*, Belo Horizonte, Brazil
Desigualdades de Bell, uma introdução a não-localidade quântica – Quantum Mechanics Graduate Course